Neural Taint Tracking for Prompt-Injection-Resistant Agents

Current Project Progress Update

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Code repository: https://github.com/muitiiifruckt/neural-taint-tracking-thesis

1 What Has Been Done So Far

The project studies defenses against *indirect prompt injection* in tool-using LLM agents. The main idea is to add an orchestration-layer taint tracking mechanism that records the provenance of data and detects when untrusted tool outputs influence later tool calls.

The evaluation is based on the **AgentDojo** benchmark, using the `workspace` suite and the `important_instructions` attack.

So far, I have completed the first baseline experiment and started the first implementation stage:

- Set up the project repository, Python environment, configuration files, and experiment scripts.
- Prepared the workflow for running the T0 baseline and collecting results.
- Ran the **T0 baseline** without any defense:
  - 10 benign user tasks;
  - 140 security cases;
  - model: `gpt-4o-mini-2024-07-18`;
  - attack: `important_instructions`.
- Collected reproducible artifacts: `results/T0_summary.md`, `results/T0_full_traces.json`, and per-case markdown traces.
- Started **T1**, a tag-only taint tracking layer that records provenance but does not block tool calls yet.
- Wrote the T1 specification in `docs/T1_SPEC.md`.
- Implemented the initial taint tracking core in `src/taint/`: `TaintedSpan`, `TaintRegistry`, propagation helpers, trace serialization, and a passive observer skeleton.
- Added unit tests for the taint tracking core, including a synthetic lineage case from email content to `send_email`.
- Prepared a roadmap document for the T0–T4 experimental stages.

Table 1: AgentDojo workspace baseline without defense.

Metric	Value
Benign Utility	100%
Utility Under Attack	20%
Security Rate	24.3%
Targeted ASR	**75.7%**

1.1 T0 Baseline Results

These results show that the undefended agent performs well on benign tasks, but remains highly vulnerable to indirect prompt injection attacks. This baseline is the reference point for the next treatments.

2 Progress Compared to the Initial Plan

The initial project plan consisted of several experimental treatments:

Stage	Description	Status
T0	Baseline without defense	Completed
T1	Tag-only taint tracking	Partially implemented
T2	Strict gate for privileged tools	Planned
T3	Field-level gate	Planned
T4	Rule-based sanitizer	Optional / later

T0 is fully completed. This is an important part of the project because it provides the reference point for all later comparisons.

T1 is currently in progress. The specification, core data structures, propagation logic, trace format, and unit tests are already implemented. The remaining T1 work is to integrate the passive observer into the AgentDojo execution pipeline so that real benchmark runs can produce taint traces.

My current estimate is:

- T0 is 100% complete;
- T1 is around 60–70% complete;
- T2–T4 are still planned.

If the first milestone is defined as "T0 baseline + initial T1 implementation", then this milestone is approximately 55–60% complete. If the full T0–T4 experimental ladder is considered, the overall progress is approximately 25–35%.

3 Plan for the Next Two Weeks

During the next two weeks, I plan to complete T1 and start T2. The planned tasks are:

1. Integrate the passive taint observer into the AgentDojo execution pipeline.
2. Run a small smoke test and generate the first real T1 taint traces.

3. Run T1 on the same benchmark configuration as T0 and verify that the tag-only layer does not change the agent's behavior.
4. Add a `taint leakage rate` metric to measure how often untrusted data reaches tool-call arguments.
5. Define the list of `privileged tools` and `sensitive fields` for the next stage.
6. Implement the initial strict gate policy for T2.
7. Run the first T2 evaluation and compare it against T0 using Targeted ASR, Security Rate, Benign Utility, and Utility Under Attack.
8. Update the written report and project documentation with the T1 implementation details and the initial T2 design.

4 Current Conclusion

At this stage, the project has a reproducible baseline, clear evaluation metrics, and an initial implementation of the taint tracking layer. The baseline confirms the main motivation for the project: the agent without defense has high benign utility, but also a high targeted attack success rate.

The next main step is to connect the implemented T1 core to real AgentDojo runs, collect taint traces, and then use this information to implement a defensive gate in T2.